Sprint Planning

Features

We are implementing two features this sprint:

1. **GenAI Personal Trainer Chatbot** - A chat interface where users can ask an AI trainer questions and receive personalized responses based on their profile data (height, weight, fitness goals, injuries, etc.).
2. **Workout Plan and Logging Page** - A dedicated page where users can create workout plans, set daily goals, track completion, and log workouts they have done.

Everything in this document is organized around these two features. Stretch goals are listed at the end after all essential work is covered.

Feature 1: GenAI Personal Trainer Chatbot

This feature covers the full AI trainer experience, from the initial chat screen to the ongoing conversation view, as well as the expanded profile that gives the AI the context it needs to personalize responses.

Issue: [UI] Create the AI Trainer Chat Page

Difficulty: 2

This is the landing screen for the AI Trainer tab. It greets the user by name and has a text input where they can send their first message to the trainer.

Deliverables:

- Build the chat landing page with greeting text, input field, and send button
- Write UI tests

Issue: [Backend] Connect the Chat Page to AI

Difficulty: 4

Wires the chat input to VertexAI so that user messages get a real AI-generated response. The user's profile data (fitness level, goals, injuries) is passed along as context so the AI can give personalized advice.

Deliverables:

- Set up and connect to VertexAI
- Send user messages to the AI and return responses to the frontend
- Include user profile data as context in each request
- Write backend tests using mocking

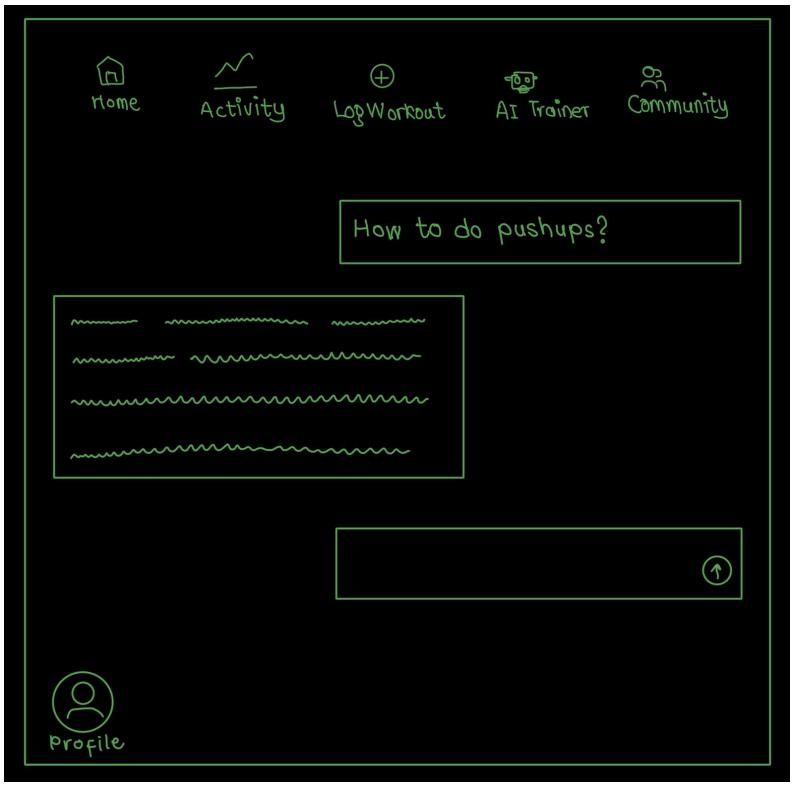
Issue: [UI] Create the Chat Thread View

Difficulty: 3

After the user sends their first message, the view transitions into a scrollable conversation thread. This is where the back-and-forth between the user and the AI trainer lives.

Deliverables:

- Build a reusable chat bubble component for both user and AI messages
- Display conversation history as a scrollable list of bubbles
- Keep a persistent input bar at the bottom for follow-up messages
- Write UI tests for the chat bubble component



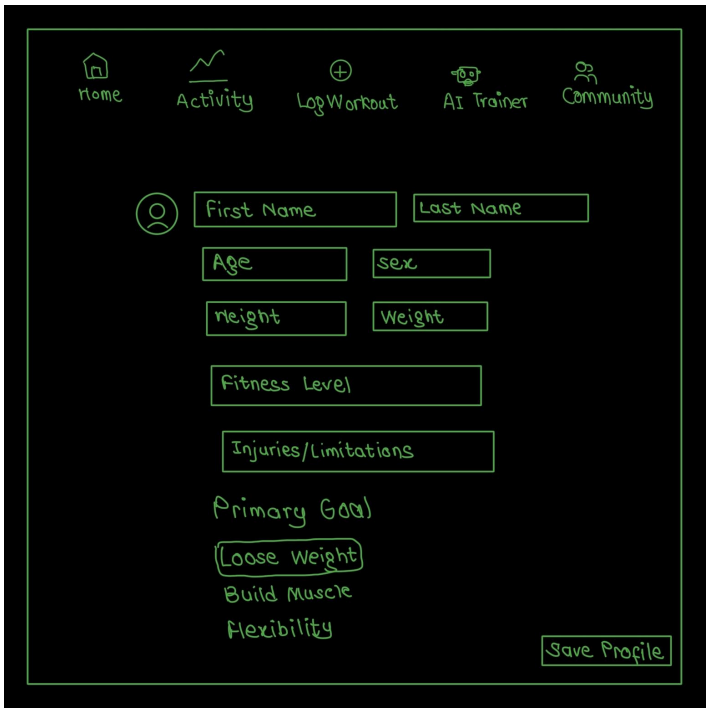
Issue: [UI] Create the Edit Profile Page

Difficulty: 2

The profile page is expanded with new fields the AI trainer uses to personalize its responses. Users can update their info and save it.

Deliverables:

- Build the form with fields for first name, last name, age, sex, height, weight, fitness level, injuries/limitations, and primary goal (Lose Weight, Build Muscle, Flexibility)
- Add a Save Profile button
- Attach a screenshot of the completed page
- Write UI tests



Issue: [Database] Update the Profile Schema

Difficulty: 1

Update the database to store the new profile fields that the AI trainer depends on.

Deliverables:

- Add the new fields to the profile table and run the migration

Issue: [Backend] Connect Profile Save to the Database

Difficulty: 1

Wire the Save Profile button so that changes actually get saved.

Deliverables:

- Create or update the API endpoint for saving profile data
- Validate inputs and return a success or error response to the frontend

Feature 2: Workout Plan and Logging Page

This feature covers everything on the workout side of the app. Users can set daily fitness goals, build workout plans, start a workout to see its details, and log workouts they have completed.

Issue: [Database] Design and Create the Daily Goals Table

Difficulty: 1

Create the database table that stores a user's daily goals.

Deliverables:

- Design and create the daily goals table (includes goal name, duration, completion status, date, and user reference)

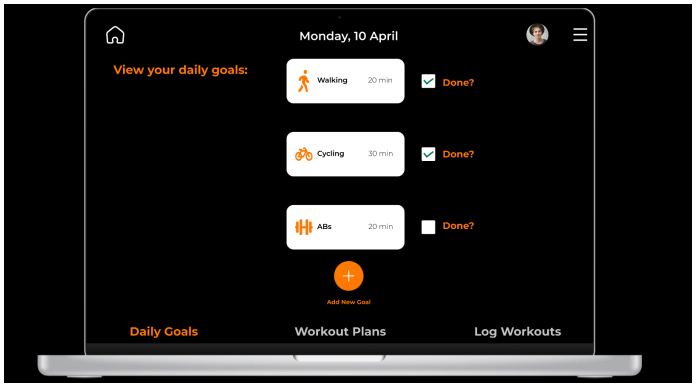
Issue: [UI] Create the Daily Goals Page

Difficulty: 2

Displays the user's daily goals for the current day. Each goal shows the name, duration, and a checkbox to mark it as done.

Deliverables:

- Build the goals list with goal name, duration, and a Done checkbox
- Show the current date in the header
- Write UI tests



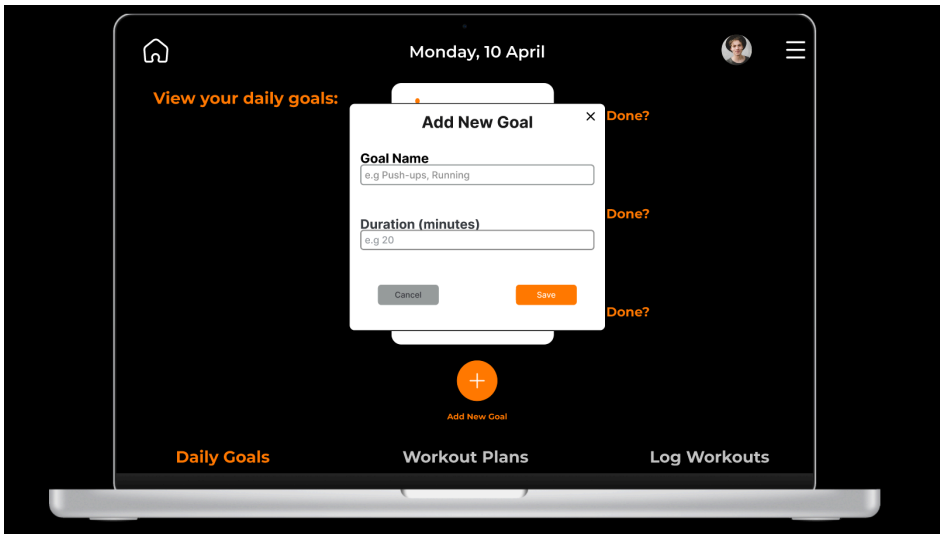
Issue: [UI] Popup for Adding a New Goal

Difficulty: 1

A modal that lets users add a new daily goal from the Daily Goals page.

Deliverables:

- Build the popup with a Goal Name field, Duration field, and Cancel/Save buttons
- Attach a screenshot
- Write UI tests



Issue: [Backend] Add New Goal and Save Completion Status

Difficulty: 2

Handles saving new goals and updating whether a goal has been completed.

Deliverables:

- Create an endpoint to save a new goal
- Create an endpoint to update the completion status of a goal

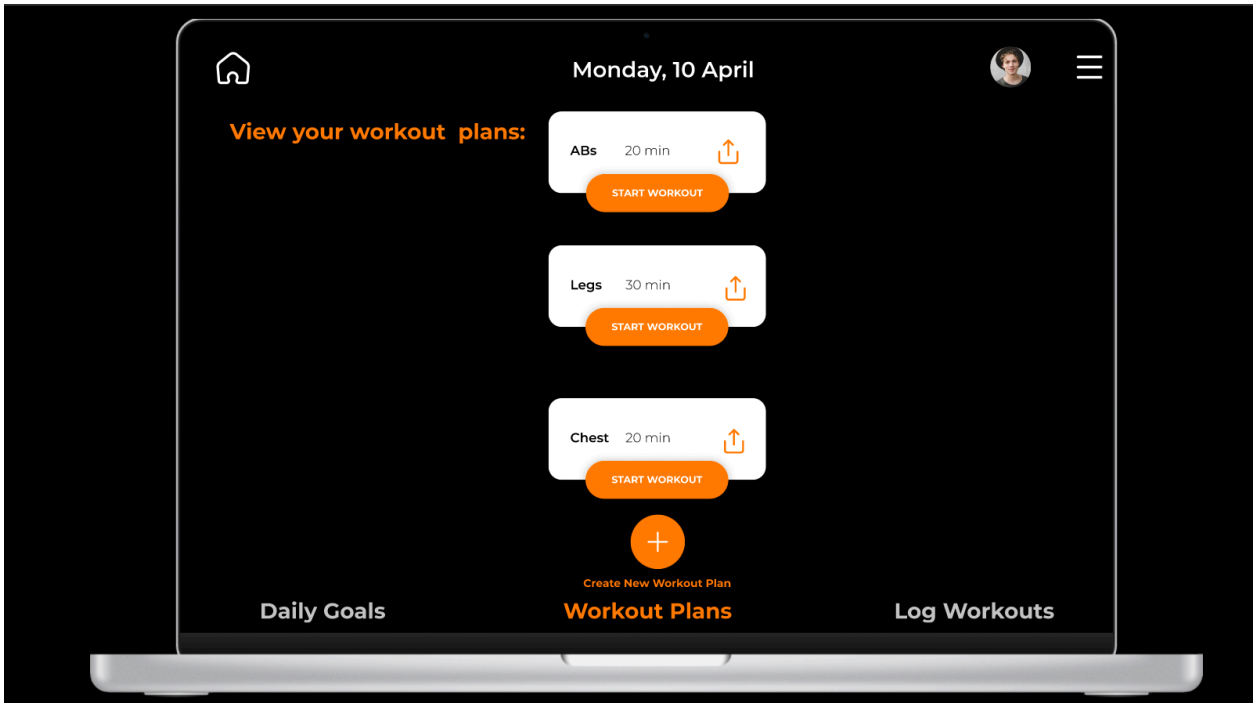
Issue: [UI] Workout Plans Page and Start Workout View

Difficulty: 1

Displays the user's saved workout plans. When the user taps Start Workout, a popup shows the workout details.

Deliverables:

- Build the workout list view with workout name, duration, and a Start Workout button
- Build the workout detail popup showing the exercise breakdown



Issue: [Backend] Fetch Workout Details on Start

Difficulty: 1

When a user taps Start Workout, the app fetches that workout's details from the database.

Deliverables:

- Create an endpoint to fetch a workout by ID and return its name, duration, and exercise list

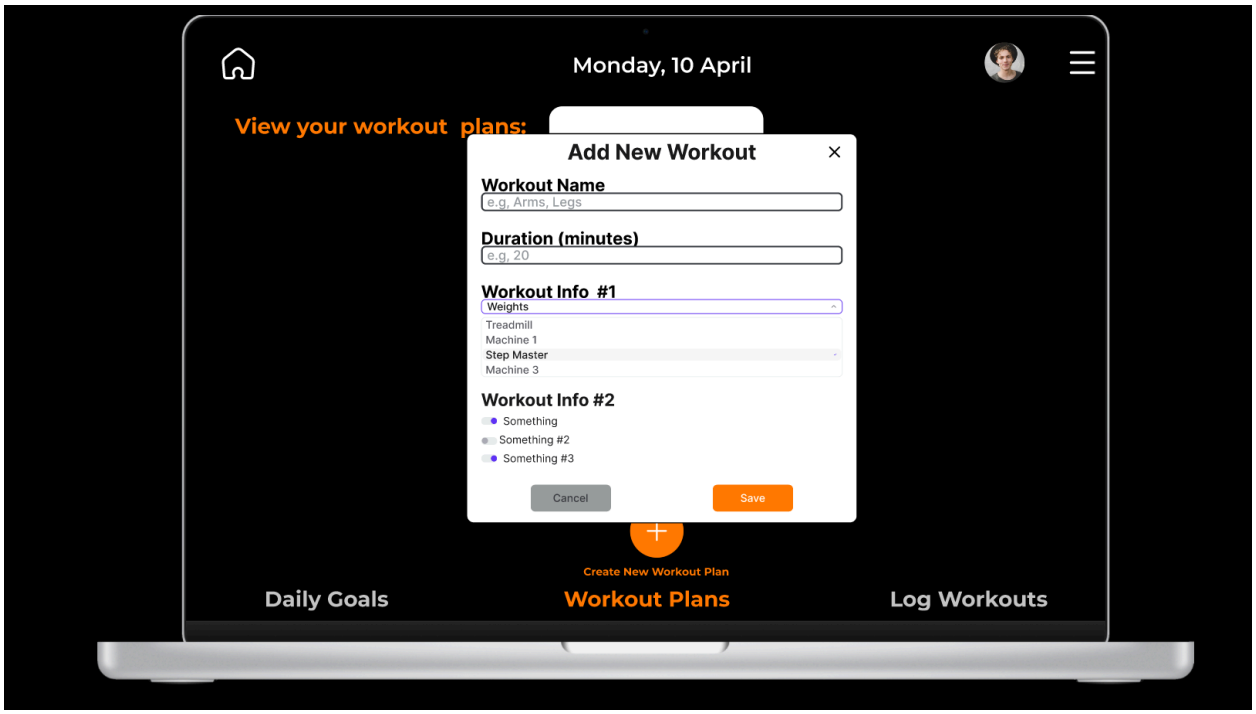
Issue: [UI] Popup for Creating a New Workout Plan

Difficulty: 1

A modal that lets users create a new workout plan.

Deliverables:

- Build the popup with fields for workout name, duration, and exercise information
- Attach a screenshot
- Write UI tests



Issue: [Backend] Save a New Workout Plan

Difficulty: 2

Handles saving a new workout plan when the user fills out and submits the form.

Deliverables:

- Create an endpoint to save a new workout plan and return the updated list to the frontend

Stretch Goals

These are lower priority and will only be picked up once all essential issues above are closed. They are still tracked as GitHub issues but are not part of the core sprint commitment.

Issue: [Backend] Save AI-Generated Content to a Workout

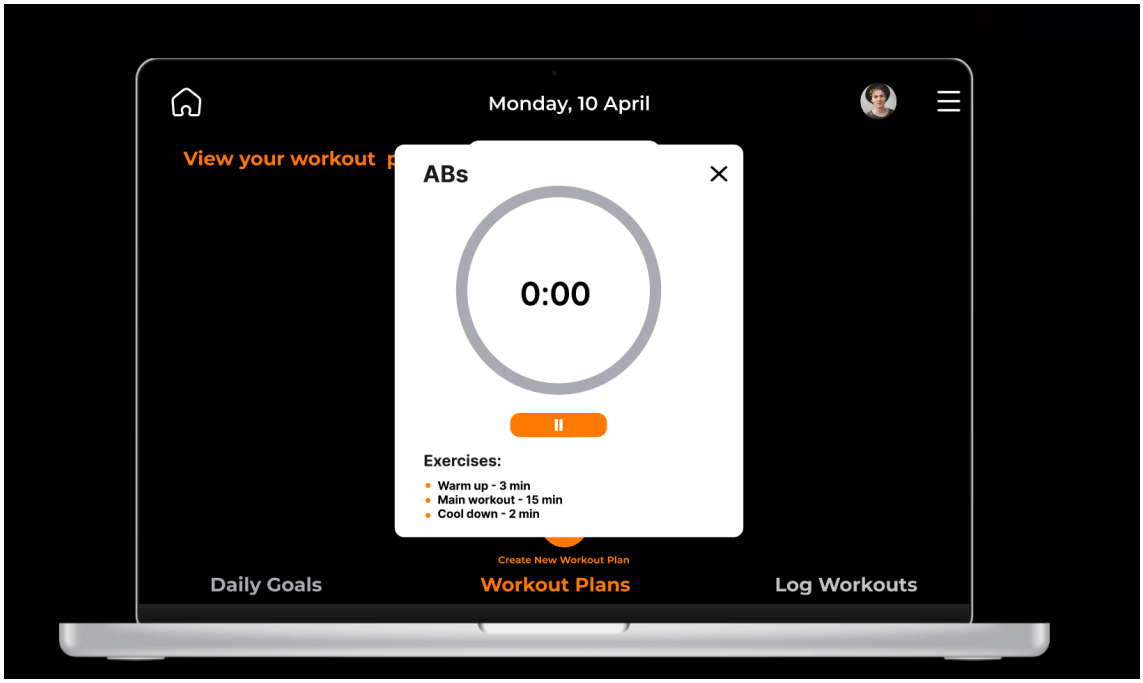
The user can highlight parts of the AI response and save them to a workout plan. Includes a Save to Workout button that appears on selection.

Issue: [Backend] Show Completed Goals on the Activity Page

When a daily goal is marked as done, it shows up on the Activity page as a recent activity entry.

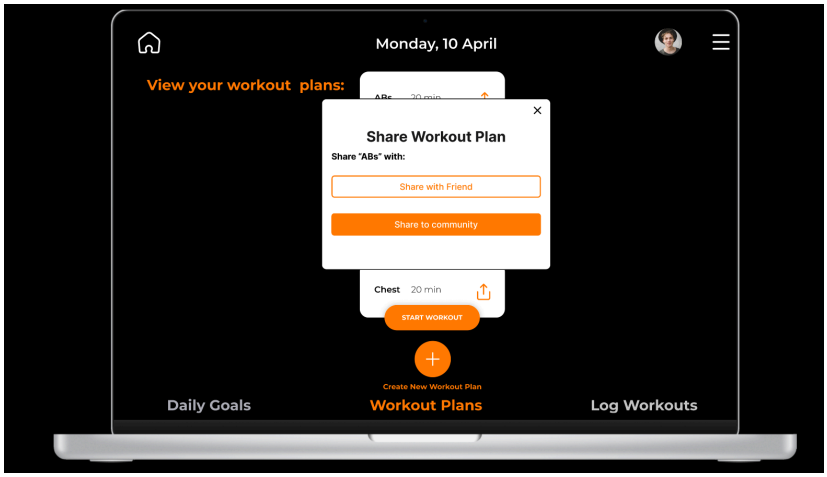
Issue: [UI] Workout Timer

A countdown or count-up timer shown during an active workout session.



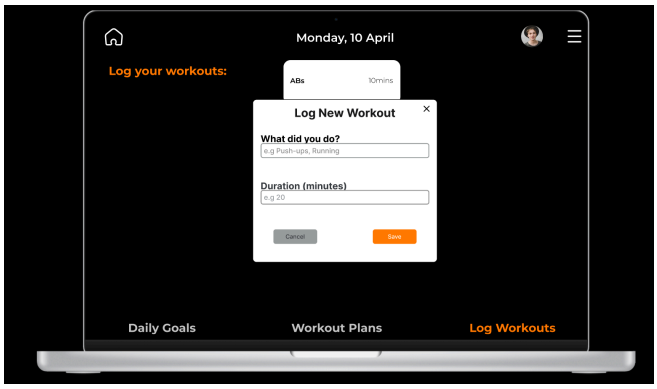
Issue: [UI + Backend] Share Workout to Community

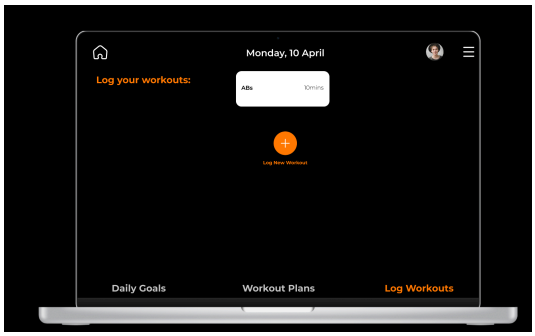
A Share button on each workout card that lets the user post the workout to the community page or share it with a friend.



Issue: [UI + Backend] Log Workouts

A Log Workouts tab where users can record a completed workout by entering what they did and how long it took.





Week by Week Plan

Deadline: April 21

We have roughly two weeks. The goal for week one is to get the database schemas set up and the core UI pages built so that week two can focus entirely on connecting everything together, writing tests, and closing out all issues before the deadline.

Week 1 (April 8 to April 14) - Foundation and Core Pages

The priority this week is getting the database ready and the main screens built so backend integration has something to connect to.

- Design and create the Daily Goals database table
- Update the Profile database schema
- Build the Edit Profile page
- Build the AI Trainer chat landing page
- Build the Daily Goals page
- Build the Workout Plans page and Start Workout popup

By the end of this week the core screens should be done and the database should be ready for integration.

Week 2 (April 15 to April 21) - Integration, Backend, and Testing

This week is for wiring the frontend to the backend, completing all remaining issues, and making sure everything is tested and working end to end.

- Connect Profile Save to the database
- Connect the AI Trainer chat to VertexAI
- Build the Chat Thread view
- Add New Goal endpoint and completion status update
- Save New Workout Plan endpoint
- Fetch Workout Details on Start
- Build remaining popups (New Goal, New Workout)
- Write all outstanding tests
- Close all issues and create the v5 GitHub release

If any time is left after the required issues are closed, stretch goals can be picked up starting with whichever is closest to done.